



SKARN

Shared Knowledge of Advanced Research Nodes

SXM-4 PCIe_{x16} Signal and Power Structure

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Introduction:

This document will provide a basic understand of the SXM-4 Mezzanine socket for Nvidia's Ampere Architecture release in mid 2020. The SXM-4 socket is a successor to the SXM-2 P100 (Pascal) & V100 (Volta)& A100 (Ampere- Autonomous Automotive Model no Nvlink). Where the SXM-3 V100 (Volta+ TDP) was an incremental step. The SXM-3 socket yielded power & NVLink expansion capability for up to 12 V100 Nodes by utilizing Nvlink switches. Ampere exceeded Volta in every way possible, down to the Sparse Computational support. Or as NVIDIA would say "Fine-grained Structured Sparsity" – (Section 3 NVIDIA DGX A100 System Architecture WP-10083-001_v01 | 3)

SXM4 features an Amphenol M-Series VR that is a 350 pin connector rated PAM4 (56gb/s) with 85 ohm impedance. This connector not only improves contact size and server installation durability but also overall signal integrity for PCIe4.0x16 and power. Now I want to pause here. **(Intuition from Chinese adapter examples say 48V but there is a chance it can also be 12V. Further testing required)**

Mezzanine
Products

Amphenol

M-Series™ VR

56GB/S HIGH SPEED MEZZANINE BGA CONNECTOR SYSTEM

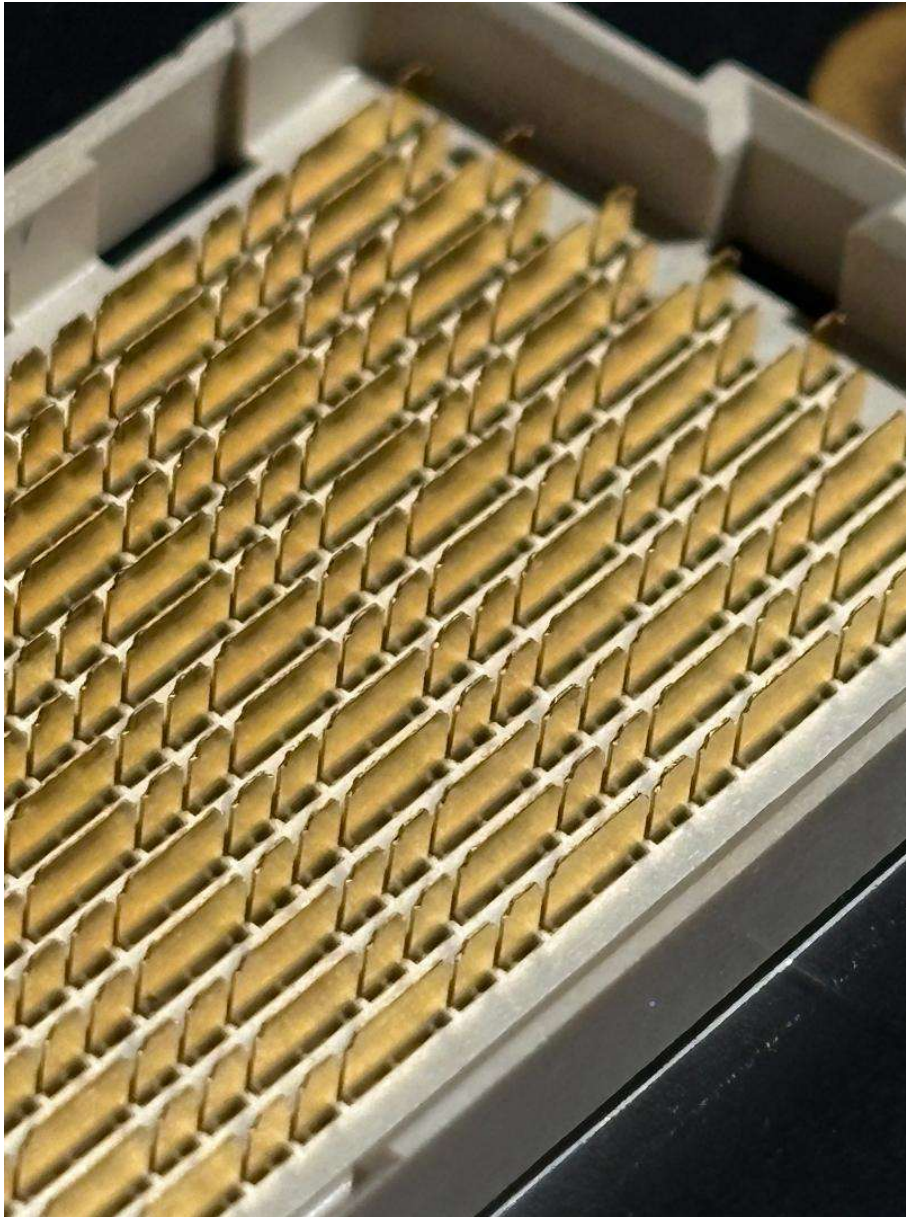
Leveraging proven technologies, including an industry-leading BGA design, this connector system offers a solution that provides superior self-aligning and self-leveling capabilities. This product incorporates a next-generation differential pair contact design, enabling impressive performance of up to 56Gb/s for both PAM4 and NRZ applications.

- High-density, high-speed, dual-point contacts
- Self-aligning/leveling for multiple connector use
- 1.27mm row to row pitch
- 56Gb/s PAM4 and NRZ
- Stack Height: 4mm

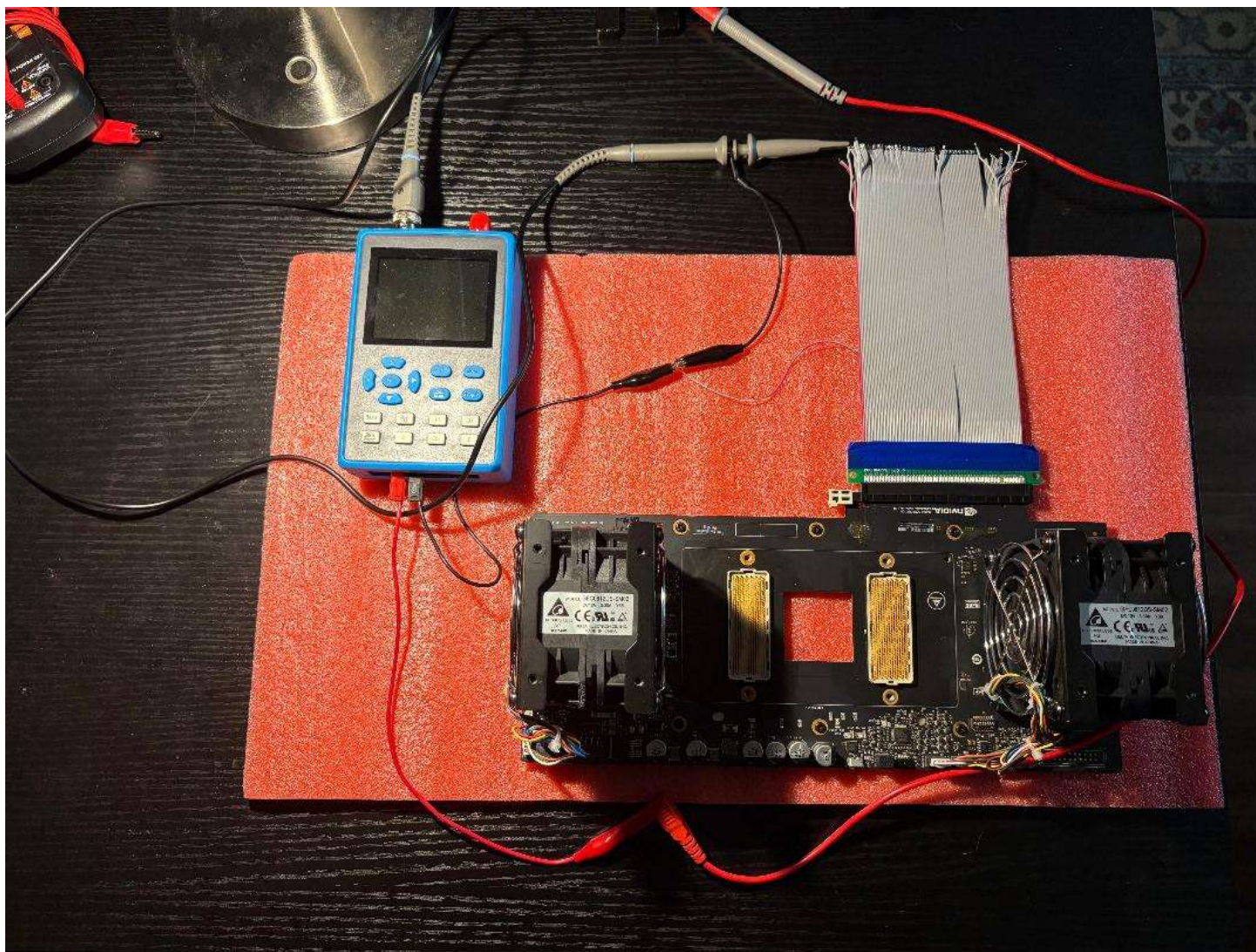


TARGET MARKETS





Close up image of SXM4 Mezzanine connector.
Note: The continuity testing was done with a cheap oscilloscope on a Chinese produced single PCIe lane adapter for the SXM4 socket. An oscilloscope was utilized to bypass DC blocking capacitors. Digital Multi-Meter was used to map out ground and power delivery pins.



SXM-4 and Beyond.....

When it comes to the ExaMAX connectors found on the Ampere and Hopper host boards. These boards usually output non-truncated PCIe4.0 and 5.0 x16 for each SXM GPU on the board. Thus we either will need to figure out the latest PCIe generation switching IC's or... we can just not solder all the pins. Thus we can have 8x H100's that offer 128PCIe lanes, and **only need to use 64 if we soldered just enough pins to allow for PCIe x8 for each GPU. Most of the communication happens between individual GPU's rather than as frequently as the CPU.** Theoretically it would be advised to use a Dual Epyc or LGA4677 system to accommodate the full 128 PCIe lanes for 8 gpu's and also peripheral PCIe necessities like NVMe storage. These trays typically utilize a Server mid-plane board that hosts several PCIe PEX or Switchtec switching chips that data multiplex bi-directionally at PCIe4.0x16 and 5.0 bandwidth. That is currently possible to replicate with modern commercial print order pcb's that adhere to impedance and signal integrity standards. This actively done in Chinese markets in the form of a single PCIeX16 expanding out to several NVMe drives Essentially sacrificing some PCIe lanes to have PCIe lanes for more M.2 storage and any additional PCIe cards....or more SXM trays.

This is still a theoretically straightforward solution to a real compute accessibility problem. This is a realistic and affordable option to get a depreciated 8x A100 80gb (640Gb Unified VRAM) fully loaded tray working with almost any motherboard if you have 54-year-old middle management money here in early 2026.

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